

Mini-Project #1. Physical Characteristics of Materials Report

Omega- Group 10

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Guide for writing a report

<https://emedia.rmit.edu.au/learninglab/content/writing-report>

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Introduction

The aim of this project is to identify the material of each given coupon by observing the stress-strain relationship of that coupon being subjected to a tensile load. Each coupon of a respective material has its own behavior when applied force (elongation, maximum force to break). The experiment is conducted in a laboratory with three unknown material coupons, plastic or metal. Through the stress-strain diagram collected from the experiment data, the material of four coupons can be specified.

* Hardware:

- Stress-Strain Apparatus: Platform to conduct experiment, stabilize coupons and pull the coupon.
- Force Sensor: record the data of the tensile force applied on the coupon.
- Rotary Motion Sensor: record the data in the rotational motion of the crank.
- Clamps: hold the coupons tightly at two ends.
- Sparklink Air: shift the data recorded from the sensors to SparkVUE app to process the data.
- Screwdriver: fix the sensors and other objects to the platform tightly for the experiment.

* Software:

- SparkVUE app: Take the data collected from the sensors and plot a force-position graph.
- Excel: Process the collected data for further calculations and plot the stress-strain graph.

Methodology:

* Stress-Strain relationship:

When a load P is applied to a bar or a coupon through the centroid of its cross-sectional area, the coupon will deform uniformly throughout the central region of its length, provided that the material of the coupon is both homogeneous and isotropic. Because the material undergoes a uniform deformation, the cross section is subjected to a constant normal stress distribution, which is calculated by the ratio of the force P applied to the cross-sectional area of the coupon

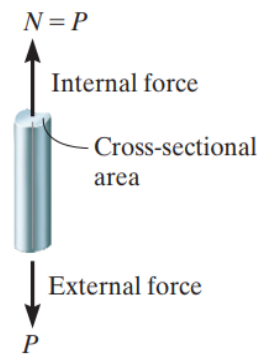
$$\sigma = \frac{N}{A} \quad (1.1)$$

Here

σ = average normal stress at any point on the cross-sectional area (N/m^2 or Pa)

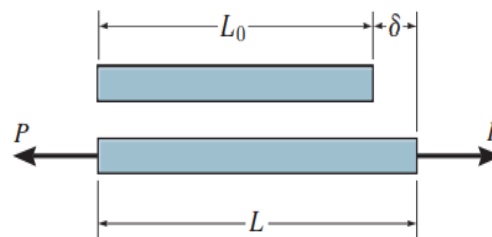
N = internal resultant normal force, which acts through the centroid of the cross-sectional area (N)

A = cross-sectional area of the coupon where σ is determined (m^2)



2. 1 - A load P is applied onto the cross-sectional area A

The deformation of the coupon refers to the change in the body's shape and size, which may be visible by the material cracks in every test. In order to describe the deformation of a body by changes in the lengths of line segments, we use strain, which is obtained by experiment:



2. 2 - A coupon is elongated by a load P

If an axial load P is applied to the coupon as in Fig. 2.2, it will change the bar's length L_0 to a length L . We will define the average normal strain ϵ of the coupon as the change in its length $\delta = L - L_0$ divided by its original length, that is

$$\epsilon_{avg} = \frac{L - L_0}{L_0} = \frac{\delta}{L_0} \quad (1.2)$$

Here

ϵ_{avg} = the average strain of the coupon (dimensionless or mm/mm)

δ = change in length (m)

L_0 = original length of the coupon (m)

L = length after elongation of the coupon (m)

Hooke's Law states that the deforming force or load a spring can exert is proportional to the distance it can be stretched or compressed: $F = kx$. This law can not only be applied to spring but also any elastic material. Thus, within the elastic region of any material, it states that:

$$\sigma = E * \epsilon$$

Where:

E = Modulus of Elasticity (Pa or N/m^2)

ϵ = The strain of the coupon on elastic region (dimensionless or mm/mm)

σ = The normal stress of the coupon on elastic region (Pa or N/m^2)

* Unknowns:

- Applied force to calculate the stress values.
- Linear displacement to define the strain values.
- Especially the material of each coupon. Thus we don't know if the coupons are ductile or brittle
- The properties and parameters of each coupon such as: Modulus of Elasticity, Yield Strength, Ultimate Strength, Tensile Strength, Percentage of Tensile Elongation.

* Knowns:

- The given theoretical material data from PASCO to calculate the unknown data as well as identify the material we are assigned to work with.

AP-8222 Plastic Test Coupons

Item	HIPS	nylon 6 (fiber)	ABS	polypropylene
Tensile strength	52 MPa	63 MPa	44 MPa	34 MPa
Experimental Modulus of elasticity	885 MPa	980 MPa	950 MPa	480 MPa
Cross Sectional Area	2.482 mm ²	2.482 mm ²	2.482 mm ²	2.482 mm ²
Tensile elongation at break	100-200%	12-16%	80-110%	100 - 650%

- Nominal gage length of the plastic test coupon is 18 mm.

AP-8223 Metal Test Coupons

Item	cold-rolled steel	annealed steel	aluminum	brass
Cross-sectional area	0.303 mm ²	0.303 mm ²	0.303 mm ²	0.303 mm ²
Tensile strength	820 MPa	720 MPa	170 MPa	430 MPa
Tensile elongation	none	3%	2%	15%
Experimental Modulus of elasticity	100 GPa	100 GPa	32 GPa	60 GPa

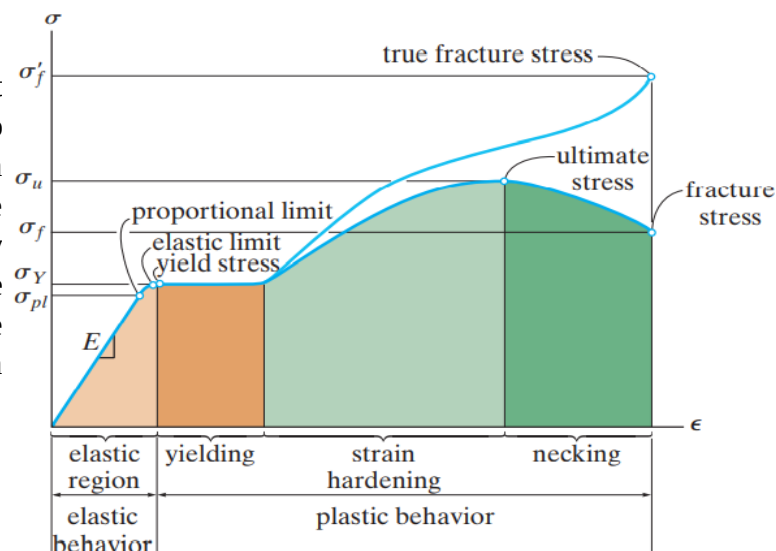
- Nominal length of the narrow section of the test coupon is 80 mm.

* Potential errors:

- The collected data could be incorrect if we forget to set the coupon in the condition that it deals no force before processing. To spot the error, we must remember to adjust the crank until the force displayed on the screen is zero.
- The applied force can be varied because the crank is turned by hand. Moreover, the clamps can loosen easily if we do not tighten them carefully, which may make the coupon break early and near the end, which ultimately affects both stress and strain data collection. To spot the error, we need to keep the clamps fixed carefully and tighten them as hard as possible, so the coupons will not slip out and break, which makes the collected data wrong.
- The applied force may not increase linearly as theory because we do not rotate the crank constantly, we sometimes stop or rotate inversely due to its reaction force. As a result, there might be some data point where the force decreases, which makes the graph appears more peaks than usual, that is against the theoretical definition. To spot the error, we have to try to rotate the crank evenly in the clockwise direction to have the appropriate data.

* Assumptions:

While doing the lab experiments with these test coupons, the Sparklink Air is connected to Element via Bluetooth, the data is illustrated on the screen of SparkVUE app, which are Force and Displacement. These variables are directly proportional to the stress and strain. Hence, we can expect Force-Displacement curve to have the same shape as Stress-Strain diagram shown in Fig. 2.3



2. 3 - Stress-strain diagram

- In this experiment, since the material of group test coupons is brittle (easy to break), the Force-Displacement curve is expected to resemble elastic region and yielding region of the stress-strain graph. The initial region of the curve resembles elastic region, which is a straight line up to a point before yielding. In this region, any increase in force will cause a proportional increase in displacement. In the yielding curve, the coupon continues to elongate without any increase in load. Any further strain increases after yielding will cause the brittle material coupon to break shortly after. However, in case of a ductile material, which can be subjected to large strains before breaking, the force-displacement diagram can have a strain hardening

curve and necking. Thus, we can easily compare the graph from the collected data with the theoretical assumption one, which reduces the mistakes we may make while plotting the graph.

- In addition, the applied force should linearly increase in the elastic region. As in some cases, the collected data of force has some decreased points due to our bad manual crank rotation. Hence, the graph when we plot will be true in comparison with the theoretical one.

* Step by step guide to carry out the experiment:

Here is the step-by-step guide to carry out the data collection process of the experiment, the images of the experiment are presented in Appendix section.

1. Set up and Calibrate the Apparatus (zero any value appears on the apparatus before starting).
2. Mount the test coupon onto the apparatus.
3. Place the lever arm in the starting position. Turn the crank counterclockwise and pull the lever arm away from the Force Sensor.
4. Press the zero button on the force sensor, click Start button. Turn the crank clockwise. Starting just before the lever arm meets the Force sensor, turn the crank very slowly.
5. Plotting the force-displacement graph on SparkVUE app.
6. Turn the crank and plot data until the coupon breaks.
7. When the coupon breaks, click Stop. Rename and save the data. Remove the coupon.
8. Repeat step 2.

We will focus on the force-displacement graph at step 5 since these data will be recalculated into stress and strain. Moreover, the coupon is examined if it is broken at the middle in step 7 or not, to make sure it was not twisted.

The Eq. 1.1 can be used to calculate stress directly since force and cross-sectional area is straightforward, however, in this project, only angle is extractable from the rotational sensor. Hence, we need an extra step to calculate the linear displacement.

$$\delta = \frac{\Delta\theta \times r}{90} \quad [1]$$

Here

δ = linear displacement in length (m)

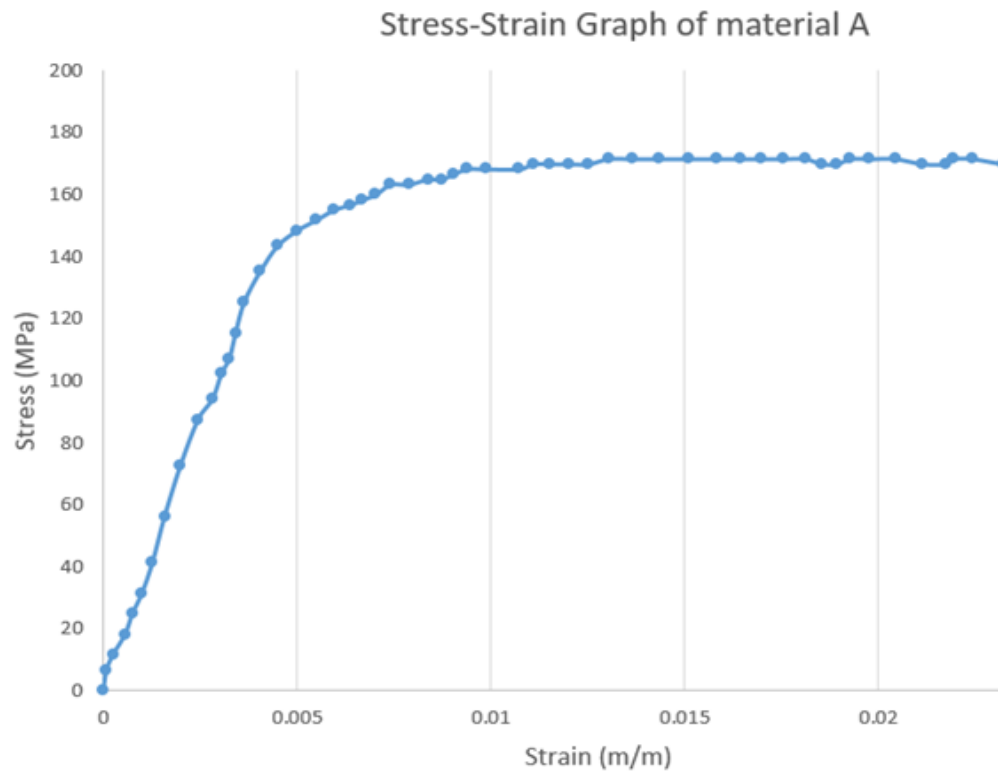
$\Delta\theta$ = collected angle from data (rad)

r = the radius of the pulley (m)

The ratio between the Rotational Motion of the sensor and the linear displacement of the coupon is 90.

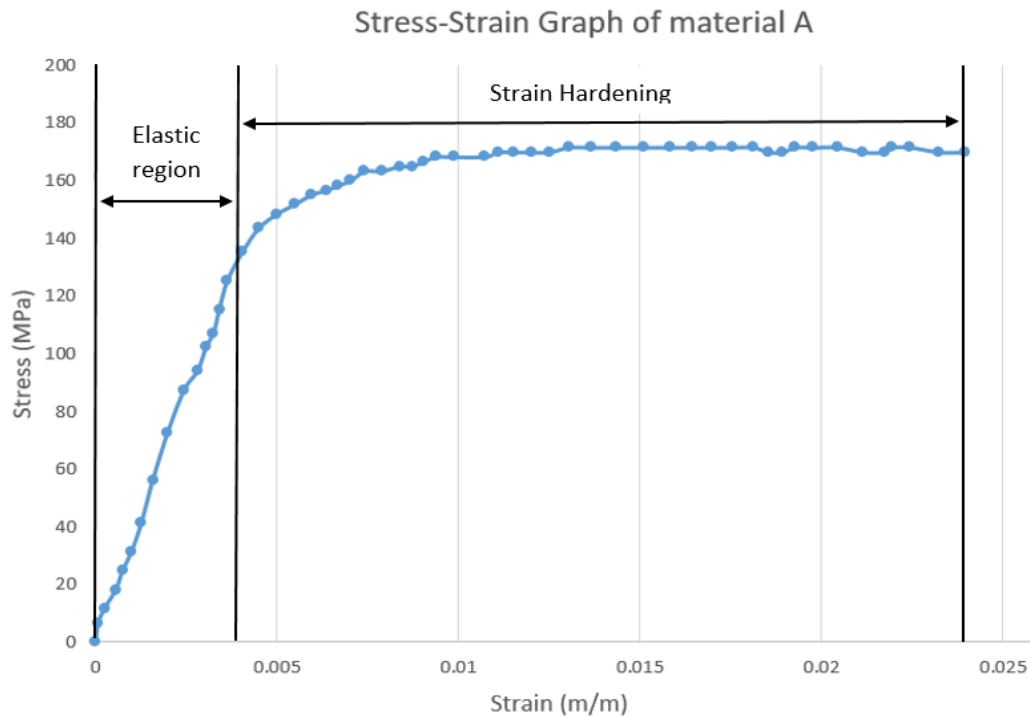
DATA ANALYSIS:

Material A:



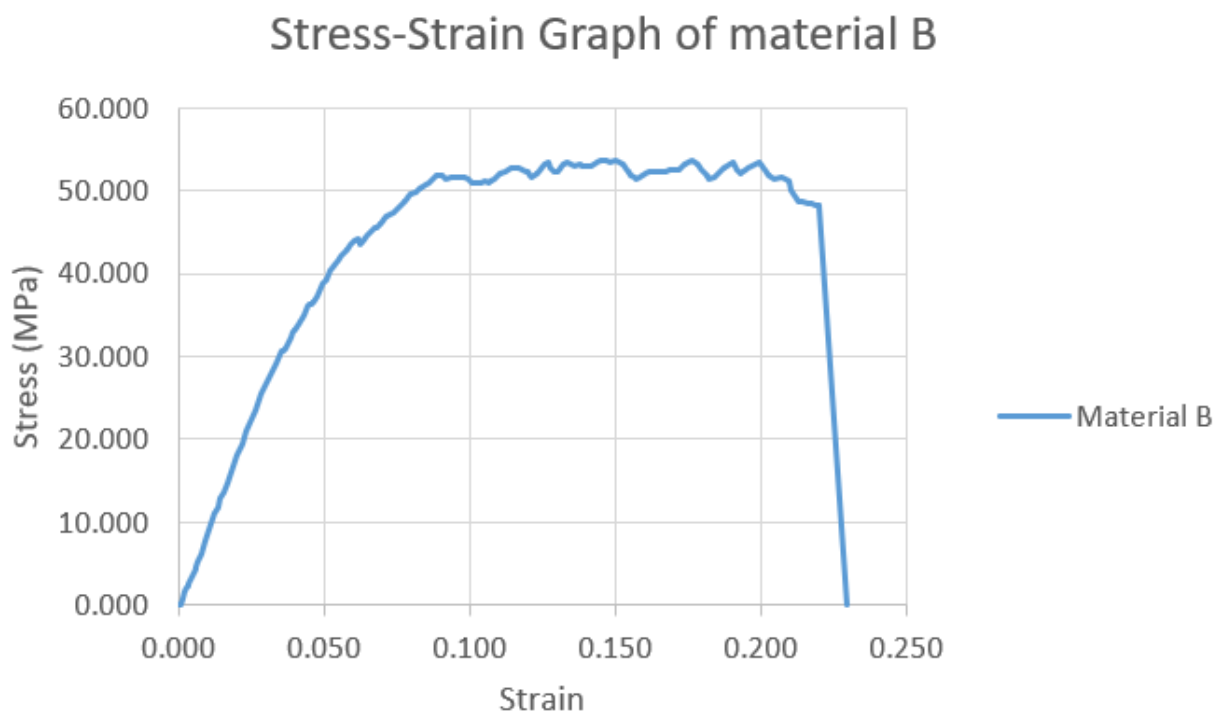
3. 1 - Stress/Strain graph of material A

According to the graph, it shows a clear elastic region as a straight line where the stress and strain increase with a constant rate of change, then there is a curve connected with a straight line which is supposed to be a strain hardening region where the stress rises a little and stays unchanged while the strain still increases.



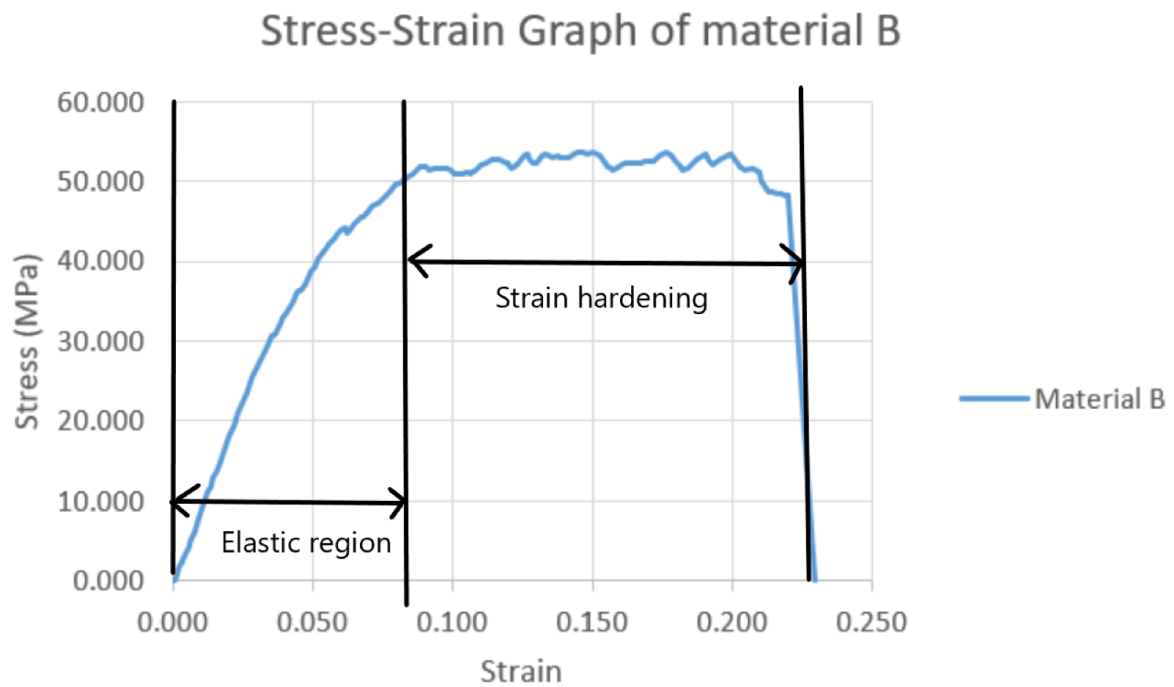
3. 2 - Region's identification of material A

Material B



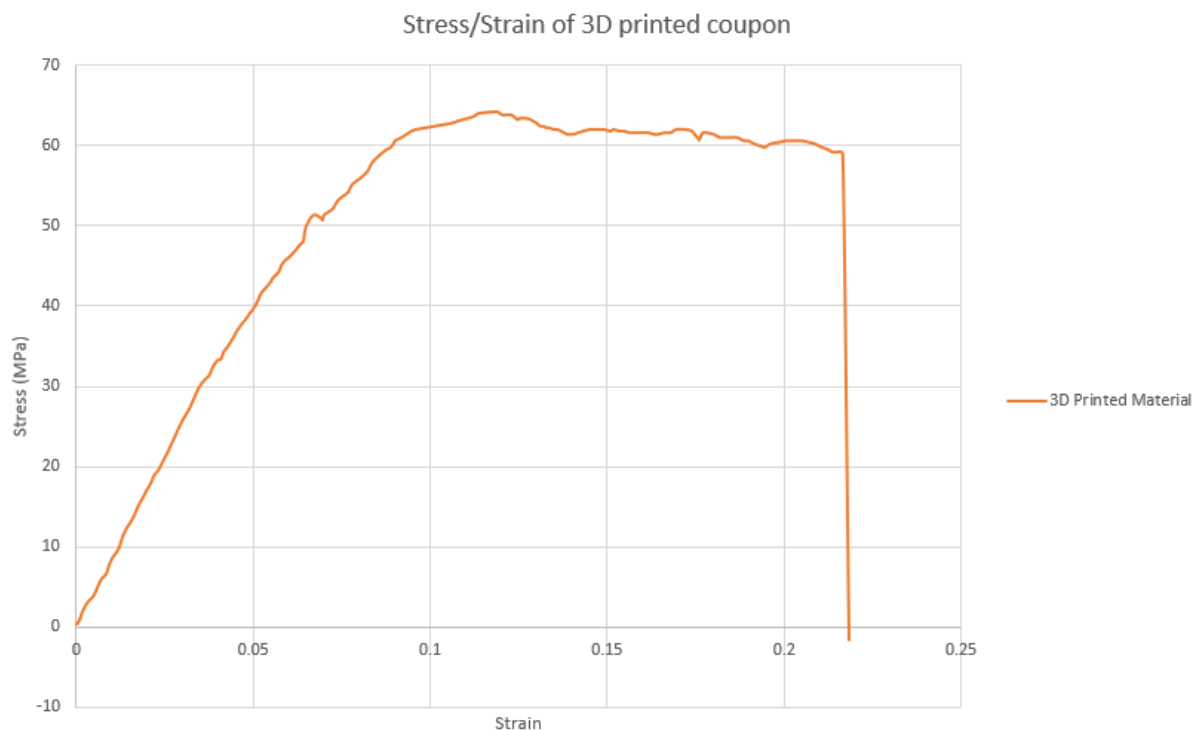
3. 3 - Stress/Strain graph of material B

From this graph it is evident that material B has very brittle properties. As the graph only shows a well-defined elastic region, followed by a very short yielding region and subsequently, fracture with little to no necking.



3. 4 - Region's identification of material B

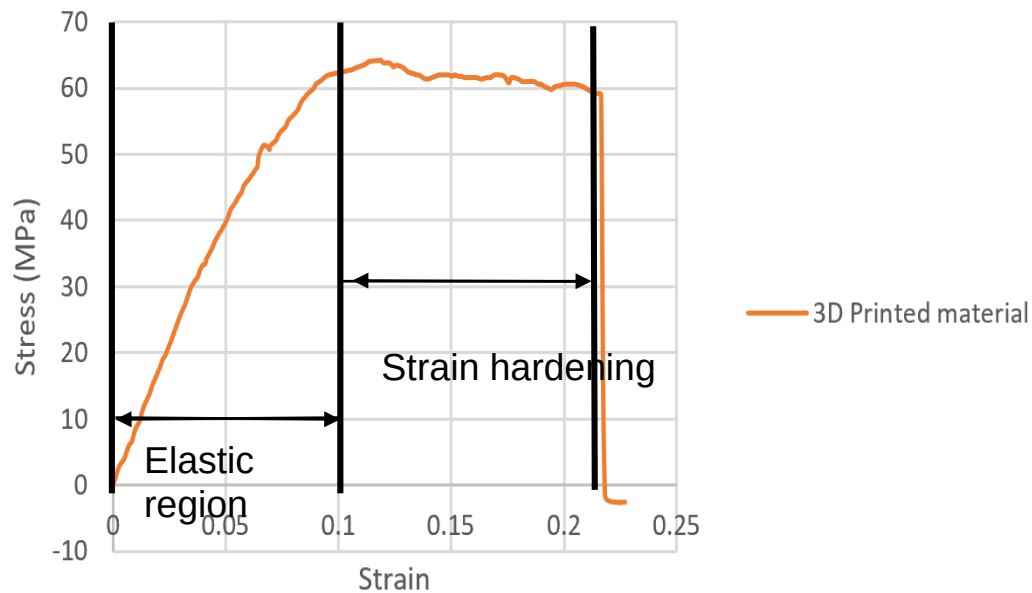
3D printed material



3. 5 - Stress/Strain graph of 3D printed material

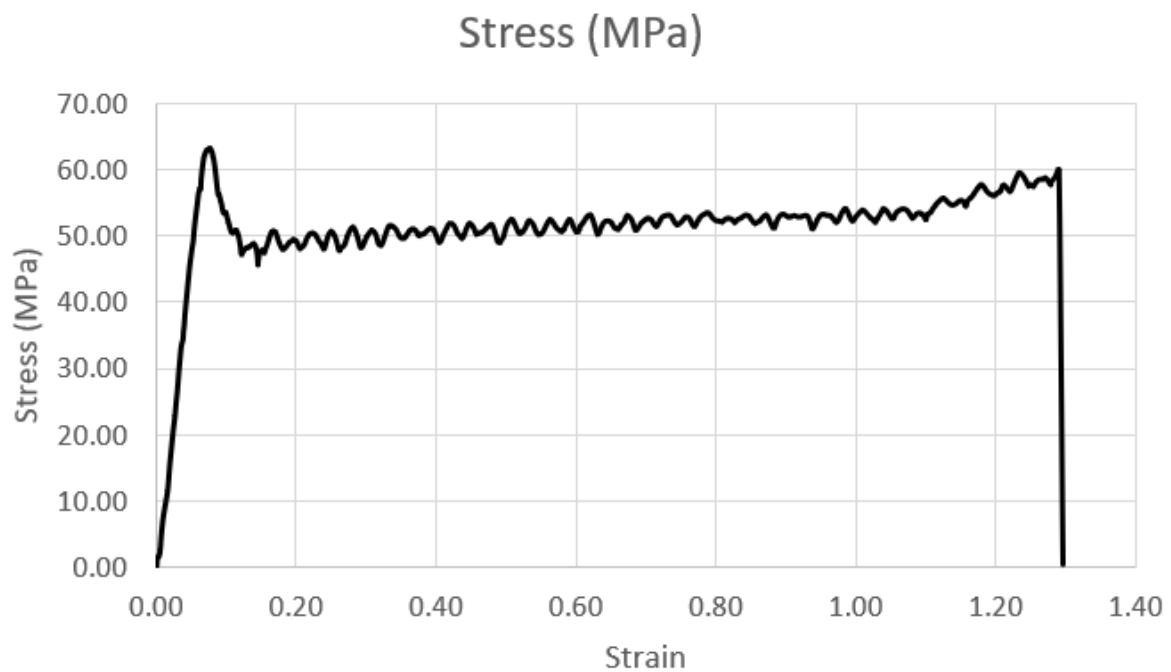
Briefly, it is evident that the stress-strain curve of this material is very similar to that of material B, being a very brittle material albeit having a slightly higher peak stress. With its regions being essentially the same as material B:

Stress-Strain Graph of 3D printed material



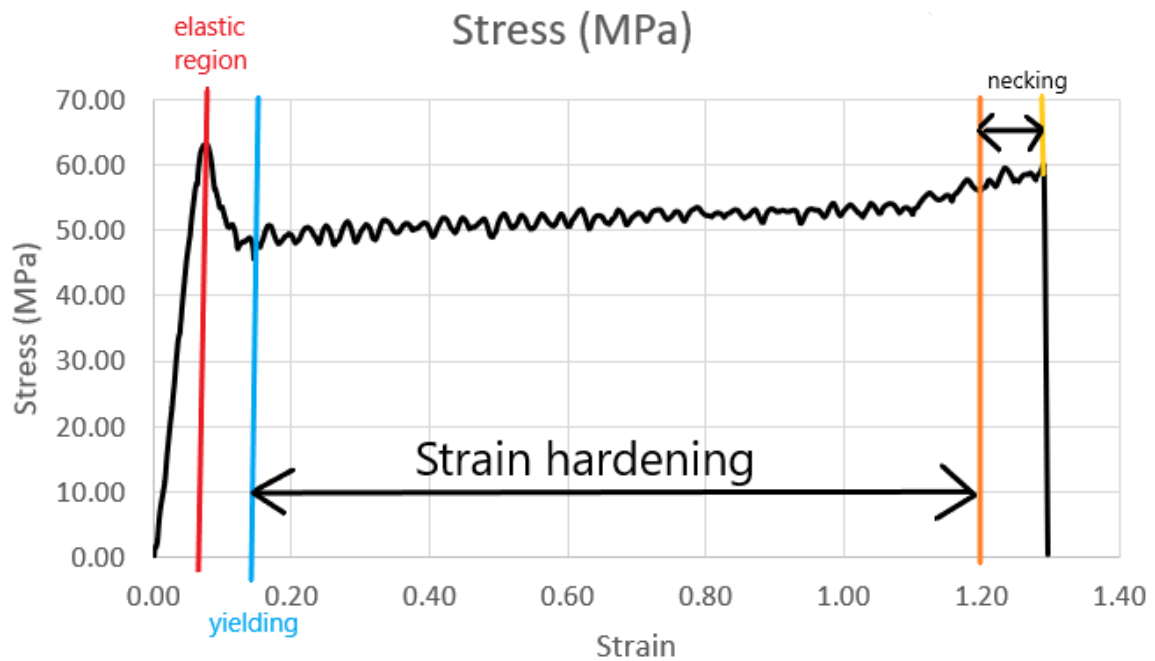
3. 6 - Region's identification of 3D printed material

Material C



3. 7 - Stress/Strain graph of material C

This graph shows that Material C shows that it possesses ductile properties. Showing a very pronounced, elastic region, yielding, strain hardening region followed by necking then fracture.



3. 8 - Region's identification of material C

* Data table and explaining calculations:

Sample	Modulus of Elasticity (MPa)	Yield Strength (MPa)	Ultimate Strength (MPa)	Tensile Strength (MPa)	Tensile Elongation (%)
A	32415.52	160.066	171.617	169.967	2.39
B	900.59	30.822	53.384	48.147	23
C	923.17	63.05	59.43	54.78	129.49
3D Printed	874.14	33.44	61.64	59.02	22.7

***Sample calculations:** Let's take the coupon A as an example to present the calculations:

- To calculate the Modulus of Elasticity, we need to find the slope of the straight line which is an elastic region on the graph. So we take the stress-strain data of two random points on the straight line for the calculation:

$$E = \frac{\sigma_2 - \sigma_1}{\varepsilon_2 - \varepsilon_1} = \frac{115.5115512 - 56.010561056}{0.00341354 - 0.0015809} = 32415.52 \text{ MPa}$$

3. 9 – Illustrated data of Stress-Strain graph

Strain	Stress
0	0
8.4583E-05	6.600660066
0.00025375	11.55115512
0.0005659	18.15181518
0.00075319	24.75247525
0.00099688	31.35313531
0.00126875	41.25412541
0.0015809	56.010561056
0.00199174	72.60726073
0.00244486	87.45874587
0.00281542	94.05940594
0.00304903	102.310231
0.00325444	107.2607261
0.00341354	115.5115512
0.00363507	125.4125413

- Yield Strength can be defined by using the 0.2 % offset method:

+ Firstly, we take the values of the strain plus 0.002 to find the offset strain.

+ Secondly, we find the offset stress by multiplying the strain values with the Modulus of Elasticity.

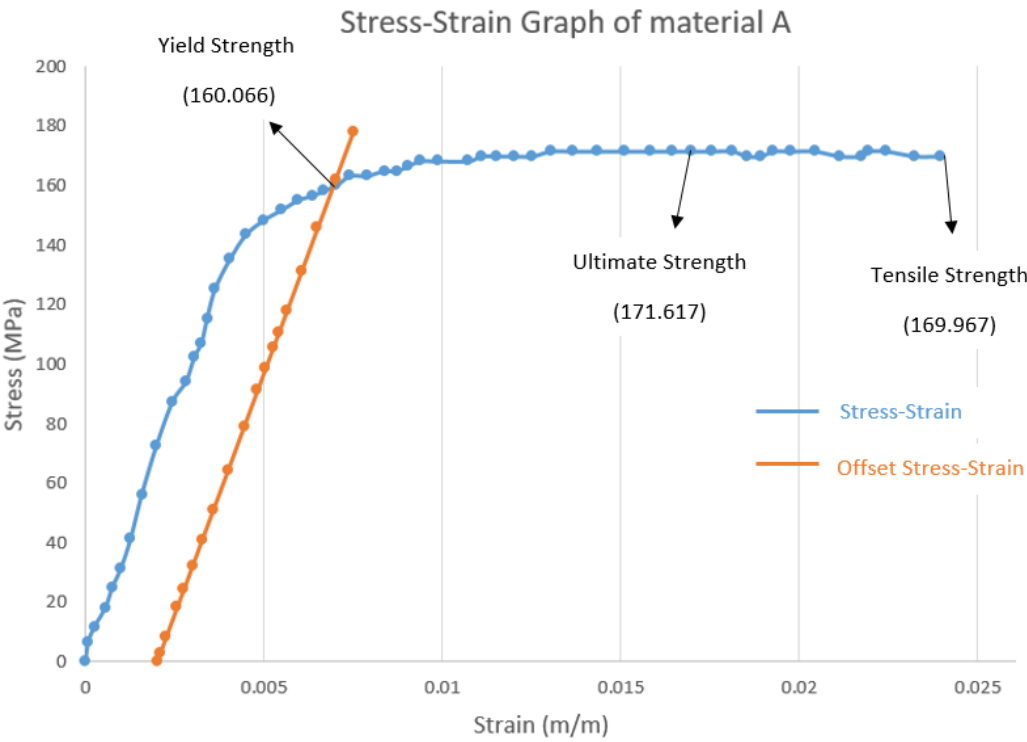
+ Finally, we draw the straight-line graph using the offset stress-strain data. The intersection point of this line and the stress-strain curve is supposed to be the yield strength. Thus, from the graph, the yield strength of coupon A is 160.066 MPa.

- Ultimate strength is the highest value of the stress on the stress-strain curve which should be 171.617 MPa.

- Tensile strength is the value of the stress shortly before the coupon is fractured and deals no stress. Hence, it should be 169.967 MPa.

- Tensile Elongation: can be calculated by taking the last linear position (elongation) and measuring its percentage from its original length (80 mm):

$$\% \text{elongation} = \frac{\delta}{L_0} \times 100 = \frac{0.001914806}{0.08} \times 100 = 2.39\%$$



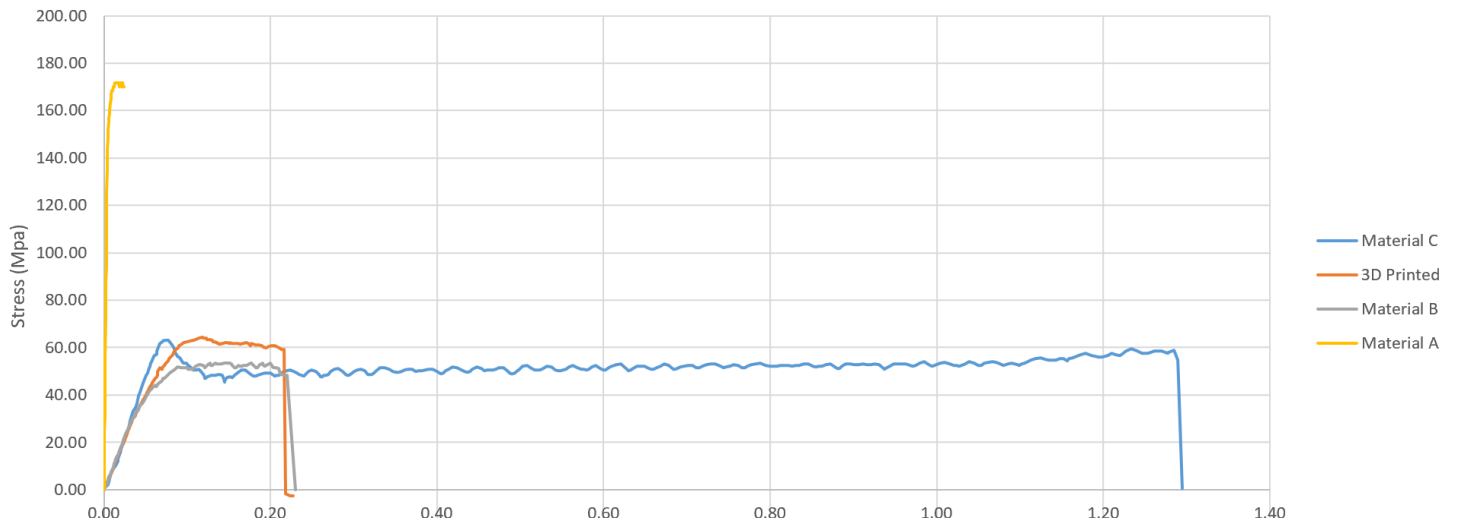
4. 1 - Finding yield strength of material A using a linear offset line

Strain	Stress	offset Strain	offset Stress
0	0	0.002	0
8.4583E-05	6.600660066	0.00208458	2.7418126
0.00025375	11.55115512	0.00225375	8.2254379
0.0005659	18.15181518	0.0025659	18.344032
0.00075319	24.75247525	0.00275319	24.415189
0.00099688	31.35313531	0.00299688	32.31422
0.00126875	41.25412541	0.00326875	41.12719
0.0015809	56.10561056	0.0035809	51.245784
0.00199174	72.60726073	0.00399174	64.56316
0.00244486	87.45874587	0.00444486	79.251442
0.00281542	94.05940594	0.00481542	91.263192
0.00304903	102.310231	0.00504903	98.835818
0.00325444	107.2607261	0.00525444	105.49451
0.00341354	115.5115512	0.00541354	110.65172
0.00363507	125.4125413	0.00563507	117.83266
0.00404792	135.3135314	0.00604792	131.21532
0.00449903	143.5643564	0.00649903	145.83832
0.00500049	148.5148515	0.00700049	162.09335
0.00549188	151.8151815	0.00749188	178.02198
0.00595104	155.1155116	0.00795104	192.9061
0.00638	156.7656766	0.00838	206.81101
0.00669014	158.4158416	0.00869014	216.86432
0.00702646	160.0660066	0.00902646	227.76629
0.00741514	163.3663366	0.00941514	240.36558
0.00788236	163.3663366	0.00988236	255.51083
0.00836972	165.0165017	0.01036972	271.30889
0.00872819	165.0165017	0.01072819	282.92895
0.00903028	166.6666667	0.01103028	292.72114

4. 2 - Data table of the offset stress strain

* Ranking materials:

Stress/Strain graph of all four materials



4. 1 - Stress Strain graph of all 4 materials

From the presented graph and the table listed above, the materials can be ranked from most ductile to most brittle as follows:

From a general view as well as practical testing, Material C is a very ductile material, having significantly longer elongation compared to the other materials, having the largest tensile elongation of 129% in Table 1.

Another conclusion can be made from this observation is that the 3D-printed coupon has a similar elastic region as well as almost having the same elongation as the material B.

Material A presents little to no yielding before failure, as shown in above figure. Here fracture of material A at 170 MPa occurred at small strain, less than 0.05 mm/mm. Hence, material A is the most brittle material. The result is shown in Table 2.

Table 2.

	Most ductile			Most brittle
Material	C	B	3D	A

*Material Identification:

From the presented data, it is time to compare the measured data with the given datasheet; With the 3 main parameters to be considered are: Young's Modulus of Elasticity, Tensile Strength, Tensile Elongation and its degree of error from the datasheet.

● **Material A: Aluminium**

	Young's Modulus of Elasticity (GPa)	Tensile Strength (MPa)	Tensile Elongation (%)
Coupon	32.41 GPa	169.967	2.39
Datasheet	32 GPa	170	2
Error(%)	1.28%	0.02%	19.5%

From this comparison table it can be inferred that the error values of coupon A's parameters ranges from 0.02% to 1.28%, with the exception of tensile elongation due to the material having little to no elongation before fracture.

● **Material B: Nylon 6**

	Young's Modulus of Elasticity (MPa)	Tensile Strength (MPa)	Tensile Elongation (%)
Coupon	900.59	48.147	23
Datasheet	980	63	12-16

Error(%)	8.1%	23.5%	43.75%
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At a glance, the data seems closer to ABS, which has a Young's Modulus and Tensile strength of 950MPa and 44 MPa, respectively. But the elongation is significantly higher, making ABS a ductile material. And with Nylon 6 being the only brittle plastic material, Nylon 6 is the chosen material for coupon B.

- **Material C: HIPS**

	Young's Modulus of Elasticity (MPa)	Tensile Strength (MPa)	Tensile Elongation (%)
Coupon	923.17	54.78	129.49
Datasheet	885	52 MPa	100-200%
Error(%)	4.3%	5.35%	0% since in range

Coupon C has produced a very accurate data set when experimented, partially due to the large amount of data that was produced from it being such a ductile material, outputting a very pronounced elastic and plastic region as well as a very accurate tensile strength value; With none of the mentioned parameter exceeding 6% in error values.

3D Printed: Nylon 6

	Young's Modulus of Elasticity (MPa)	Tensile Strength (MPa)	Tensile Elongation (%)
Coupon	874.14	59.02	22.7
Datasheet	980	63	12-16
Error (%)	10.8%	6.31%	41.9%

As seen in the presented graph, the 3D printed coupon almost has an identical elastic region to material B, as well as the same fall-off fracture point, but with a different stress value.

Conclusion

With the materials of the coupons identified using the calculated stress-strain from the post-processed data set and have concluded that the outcome of the practical data has proven to be as expected when compared to its theoretical counterparts, with material C having the highest Modulus of Elasticity (923.17 MPa) and tensile elongation (129.49%) despite material A

having significantly higher Modulus of Elasticity (32.41 GPa) but has insignificant tensile elongation (2.39%). This suggests that the Modulus of Elasticity is inversely proportional to the tensile elongation of the material. So this has sparked the question – **Can a material have a high Modulus of Elasticity (at least 70GPa) as well as a high elongation (40%)?**

First off, it is necessary to look at the formula for Modulus of Elasticity:

$$E = \sigma / \epsilon \Leftrightarrow 70 \times 10^3 = \sigma / (40\%) \Rightarrow \sigma = 50400 \text{ MPa}$$

This formula shows that, for a material to have at least 70GPa Modulus of Elasticity and 40% elongation, the stress-to-strain ratio is impossible to achieve. So, it is impossible for a material to have both high Modulus of Elasticity and high tensile elongation.

Which material has the most potential for strain hardening?

Since strain hardening is the process of increasing yield strength of a material while decreasing its ductility, making it more brittle. So, the more ductile a material is, the more potential for strain hardening it has. Hence the material to be under examined here is **HIPS**.

References

[1] R. C. Hibbeler, *Mechanics of Materials*. Hoboken: Pearson, 2022.